A quality-controlled PM2.5 dataset with frozen cross-city evaluation splits for six Central Asian cities

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Abstract

We present a quality-controlled dataset of daily PM2.5 observations from 7 instruments in 6 Central Asian cities — Almaty, Ashgabat, Bishkek, Dushanbe, Khujand and Tashkent — covering 2018-11-27 to 2024-12-31. Records were screened by seven pre-registered quality rules, plus one added after validation exposed a duplicated instrument, then deduplicated and timezone-verified; 5 are US embassy reference monitors and 2 low-cost sensors, labelled as such. Each station-day is paired with satellite retrievals, a chemistry-transport forecast and static geography, each with a measured acquisition latency fixing whether it could be known at prediction time. The release includes frozen, checksummed splits (blocked-temporal with a 240-hour purge; leave-city-out over 6 folds), a baseline ladder and reference model outputs. Predicted city means for held-out cities span 6.77 µg/m³ against 35.23 observed, so bias falls with concentration largely by construction (rho = -1.00); fold RMSE rises with it but not after normalising by each city's variability (rho = -0.03). The data support cross-city generalisation, satellite-ground fusion, low-cost-sensor transfer and reproducible benchmarking.

Background and Summary

Central Asia is among the most polluted inhabited regions on earth, and among the least instrumented. Tursumbayeva et al. (2023) put annual PM2.5 in six regional capitals at 4.3–12.6 times the WHO 2021 guideline, tracing the burden mainly to coal combustion rather than transport, against official emissions inventories. Source-apportionment studies reach the same conclusion independently in Kazakhstan (Tursun et al., 2025) and Tajikistan (Papagiannis et al., 2024). The monitoring base beneath those numbers is thin and unevenly open: Turkmenistan operates no national network, Kazakhstan releases data only to users physically inside the country, and only Kyrgyzstan publishes in a fully open form (OpenAQ, 2025).

Estimates are not what the region lacks. Global gridded products (van Donkelaar et al., 2021) already assign PM2.5 values across Central Asia, and the epidemiological literature consumes them. What nobody can do is check those estimates, or set two methods against each other on identical terms. There is no open station-level benchmark for the region — no frozen splits, no declared protocol, no shared evaluation.

The cost of that absence is measurable. Consider the closest analogue: PM2.5 estimation across Xinjiang, an arid, dust-affected, sparsely monitored region much like this one. Jin et al. (2022) report R^2 between 0.73 and 0.81 under 10-fold cross-validation with no spatial stratification. With 41 stations in 16 cities and 8-day averaging, that design places observations from the same station, often the same window, on both sides of the split. Such a figure measures interpolation among stations already held rather than estimation where no monitor exists, and nothing in a reported R^2 separates the two. The remedy is established: blocked validation for spatially, temporally or hierarchically structured data (Roberts et al., 2016), for spatially derived predictors (Meyer et al., 2019) and for particulate data specifically (Alazmi and Rakha, 2022), with validation strategy named among the systematically overlooked issues in the field (Tang et al., 2024). What the region has never had is a benchmark that enforces it.

AQ-Bench (Betancourt et al., 2021) is the precedent — 5,577 stations worldwide, split by spatial clustering at a 50 km threshold — but it targets long-term ozone metrics from station metadata in a time-independent regression, and excludes Central Asia entirely. We borrow its spatial clustering rationale rather than inventing a second one, and diverge on pollutant, target, temporal protocol and region. AirDelhi (Chauhan et al., 2023) offers a second precedent for fine-grained particulate benchmarking, confined to a single city.

This paper documents a benchmark of 7 instruments across 6 cities spanning 2018-11-27 to 2024-12-31. Splits were frozen and hashed before the reported results were produced: blocked-temporal with a purge gap of 240 hours from the maximum feature lag (168 h) and horizon (72 h), leave-city-out over 6 folds, and leave-station-out where station density allows (2 folds; Almaty, Ashgabat, Bishkek, Dushanbe, Tashkent hold one instrument each and are named ineligible rather than quietly dropped). Immutability is enforced by a test that fails for the authors exactly as for anyone else. Every predictor carries a measured latency and a typed availability flag, so anything that cannot exist at prediction time is barred from deployable configurations by test rather than convention.

One fold deserves naming. Khujand's stations begin after the training block closes, so the city contributes no training label anywhere in the record — not one row. A model arrives with no local history and must return a concentration regardless. This is the harshest test the benchmark contains, and the one that matches the deployment case the work exists for: an unmonitored city asking for a number it has never been given. We report it separately; averaging it into the other five would report neither.

The intended reuse is direct. Any method producing a PM2.5 estimate, statistical, physical or learned, can be scored on these splits against the included baseline ladder without re-deriving a protocol, and without the ambiguity that makes existing regional figures incommensurable.

Methods

Study region and period

The benchmark covers 6 Central Asian cities — Almaty, Ashgabat, Bishkek, Dushanbe, Khujand and Tashkent — over 2018-11-27 to 2024-12-31. Cities were not selected: they are every city in the region with an openly published PM2.5 record meeting the inclusion rules below. Turkmenistan operates no national network, and Kazakhstan releases national-network data only to users physically inside the country, so both are represented solely by US diplomatic-post monitors.

Ground observations

Hourly PM2.5 was retrieved from the OpenAQ v3 archive for all stations within the region's bounding box. 317 candidate stations were assessed. Inclusion required, as pre-registered before the data were inspected:

These are preliminary data. The reference-grade observations originate with the US EPA AirNow programme, which states that its observational data "are not fully verified or validated" and "should be considered preliminary", and that they "are not subjected to the full validation used to officially submit and certify data in EPA's regulatory database". Fully validated regulatory data are held in EPA's Air Quality System and were not used here. The quality-control suite below is therefore applied to preliminary data and does not substitute for the originating agencies' validation. Any result computed on this benchmark inherits that status.

- **Q7 span and completeness** — at least 2 years of record and 60% completeness within it. This is the rule that excludes 306 stations, almost all low-cost units, and that excludes Astana at 42.8% completeness.
- **Q1 physical range** — values outside [0, 1000] µg/m³ are masked.
- **Q2 flatlining** — ≥24 consecutive identical non-zero values are masked.
- **Q3 zero runs** — ≥6 consecutive zeros are masked.
- **Q4 unit sanity** — station median outside [1, 500] µg/m³ rejects the series, catching mg/m³ reported as µg/m³ and AQI values reported as concentrations.

Every rule records its effect on *n* and its direction of bias if wrong; the full decision log is `data/DECISIONS.md`.

Duplicate resolution (Q5). Two distinct failures are handled separately. *Q5a*: one identifier appearing at more than one coordinate rejects the series. *Q5b*: several identifiers within 150 m are one instrument, detected by single-link clustering over the haversine distance, and merged. *Q5c*: any pair whose overlapping observations are bit-identical on more than half of samples is one instrument regardless of separation. Q5c was added after Q5b passed a pair

6.06 km apart that proved to be a single US-embassy monitor republished under two programmes; the evidence is given in Technical Validation.

Merging is by **precedence and gap-fill, never averaging**: the feed with more observations is primary, the other fills only the hours the primary lacks, and per-hour provenance is retained in `panel_sources.parquet`. Averaging two copies of one measurement reduces no noise and, where the copies disagree, fabricates a third value no device produced. Three pairs merge under this rule — Bishkek, Ashgabat and Dushanbe — giving 7 instruments across 6 cities.

Q6 timezone verification. Metadata offsets are not trusted. Each station's diurnal composite is cross-correlated against a regional reference, and the check reports lag identifiability rather than asserting a shift: Central Asian urban PM2.5 is bimodal with peaks roughly half a day apart, so whole-shape correlation cannot separate a 12-hour offset from none. Where the hypothesis is not distinguishable the station is flagged, not rejected.

Daily target. The prediction target is the local-calendar daily mean, requiring at least 18 hourly observations. Local days rather than UTC: a UTC boundary splits a Central Asian night in half, cutting the overnight inversion peak across two days and understating both.

Predictor sources

Each predictor carries a **measured** acquisition latency and a typed availability flag, so that anything which cannot exist at prediction time is excluded from deployable configurations by test rather than by convention. Latency was measured rather than assumed; three of five initial estimates proved wrong, one by 774 days.

- **Satellite.** Sentinel-5P CO, NO_2, SO_2 and absorbing aerosol index (Veefkind et al., 2012); MODIS MAIAC AOD (Lyapustin et al., 2018). Retrieval-quality fractions are carried as features, because missingness in these products is correlated with the target — SO_2 retrieves on 0.1% of December days against 91.0% in July, a solar-geometry floor that blinds the direct winter coal tracer throughout the coal season. Null rows are retained rather than imputed: interpolating across a systematically missing extreme tail would invent the values that matter most.
- **Chemistry transport.** Copernicus CAMS global PM2.5 forecast, used both as a feature and, bias-corrected, as a baseline.
- **No meteorological predictor.** ERA5 single-level fields (Hersbach et al., 2020) were retrieved and latency-tested; at 163 h they cannot enter the deployable tier, and the retrieval was stopped before a multi-year archive existed, so they enter no tier. The reference model carries no boundary-layer height, wind, temperature or humidity term.
- **Static geography.** Elevation, terrain basin indices at multiple radii, VIIRS night-time lights, and population density.

Split construction

Splits were built by `src/ecopulse_ca/splits/builder.py`, hashed, and committed. They are immutable by test: `splits.sha256` is compared against a fresh build on every run, and the test fails for the authors exactly as for anyone else. Changing a split requires raising the benchmark version, recording the reason in `data/DECISIONS.md`, and regenerating every published number — deliberately more work than editing a JSON file.

Temporal blocks. Train, a purge, validation, a second purge, test, and a reserved post-test block. The test block is calendar 2024, the last full year of reference coverage in the frozen panel: every StateAir feed stops on 2025-03-04, when that publication channel closed. Each purge is 240 hours, derived rather than chosen: `purge_hours = max_lag_hours (168) + max_horizon_hours (72)`, so that no training row's feature window can reach into the block that follows. The relation is stated in `config.purge_rule` so a reader can verify it.

What that bound covers, precisely. `max_lag_hours` bounds the features admitted under leave-city-out, which is the protocol every headline number in this paper is computed under. Task F additionally admits a station's own history, including a 30-day rolling mean whose window reaches 720 h and therefore crosses the purge into the preceding block. That is deliberate and it is not leakage in the leave-city-out sense: a forecaster at a monitored station genuinely holds that station's past observations at prediction time. It does mean a Task F test row's own-history features can overlap the validation block used for tuning, so Task F figures carry a weaker separation guarantee than Task N figures and the two are never compared. A submission that uses an own-history window longer than 168 h under Task N would violate the protocol; two tests in `tests/test_purge_gap.py` pin the exclusion and the declared window length so neither can drift unnoticed.

Leave-city-out (6 folds). Each fold withholds one city entirely — every station in it, for the whole record. This is the protocol the benchmark is built around, because it is the only one that answers the deployment question: what is the concentration at a location with no local monitor?

Spatial blocking is contested, and the objection applies to a different estimand than this one. Wadoux et al. (2021) show that spatially blocked cross-validation is pessimistically biased for *map accuracy* over a fixed population, where design-based validation on a probability sample is the correct estimator. The quantity here is not map accuracy; it is the error incurred in a city that contributed no training row. For that, withholding the city is the estimand rather than a biased proxy for it, and a random split estimates something else. Meyer and Pebesma (2021) give the complementary frame: whether a held-out city falls inside the model's area of applicability. Technical Validation answers that question empirically: held-out-city bias falls monotonically with the city's mean concentration (Spearman rho = -1.00), while fold RMSE carries no such relation after normalisation by each city's own variability (rho = -0.03).

Leave-station-out (2 folds). Available only where a city holds more than one instrument. Almaty, Ashgabat, Bishkek, Dushanbe, Tashkent hold one instrument each and are named ineligible rather than quietly dropped.

Both remaining folds are the Khujand pair, and both of those instruments are low-cost sensors. Merging the two co-published Dushanbe records into one instrument (see above) removed the only reference-grade city holding more than one device, so within-city station holdout is now evaluated *exclusively* on the Clarity pair. This protocol therefore says nothing about generalisation across the reference-monitor network, and it supports no headline claim in this paper. Leave-one-out protocols carry a documented failure mode of their own (Austin et al., 2025), which is a further reason not to rest a headline number on two folds. The folds are retained because they are well defined and may be useful to a method specifically targeting low-cost sensor transfer.

Reference implementation

Three feature tiers are defined by availability at prediction time — `static_only`, `deployable` and `retrospective` — so that a retrospective number can never be mistaken for a deployment claim.

Two tasks are defined separately and never pooled. **Task N** (nowcasting at unmonitored sites) withholds whole cities and admits no local history. **Task F** (forecasting at monitored stations) permits the station's own lagged observations; as implemented here it predicts the next-day daily mean from lags of 1, 2 and 7 days plus 7- and 30-day rolling means, and is therefore single-horizon.

The reference model is LightGBM. Hyperparameters are selected by grid search over 16 combinations on the validation block only; a test parses the tuning function and fails if it references the test block at all. The selected configuration is then refit on train plus validation, with the purge block withheld, and scored once on the test block per configuration. Every configuration runs 5 seeds (0, 1, 2, 3, 4) and is reported as mean ± standard deviation.

Baselines. A mandatory ladder that every submission must climb: persistence, climatology, a training-pool constant, nearest-monitor, inverse-distance weighting, ordinary kriging, and CAMS in three variants. The CAMS bias correction is fitted on the training block only; the pooled variant excludes the held-out city from its own correction, so no label from a city informs its own reference. Baselines are additionally scored at daily resolution on the models' own evaluation rows, because an hourly RMSE is not comparable to a daily one.

Statistical methods

Model-versus-baseline comparison uses the Diebold–Mariano test on squared-error differentials with Newey–West HAC variance and the Harvey–Leybourne–Newbold small-sample correction. Because the loss differential is serially correlated within station and station-days cluster within cities, a pooled station-day test is not a valid inference; the primary analysis aggregates to one value per city and tests those 6 observations, with sensitivity analyses across HAC truncation lags and a cluster bootstrap. With 6 clusters an exact sign-flip permutation test is reported alongside the parametric one, and its attainable floor is stated. Per-fold tests are Holm-corrected for multiplicity (Holm, 1979), and are read as descriptive diagnostics rather than as the inferential claim, since both comparators are estimated models rather than given forecasts (Diebold, 2015). Full detail and results are in Technical Validation.

Software and generative-AI assistance

Analysis code is Python 3.12; exact dependencies and versions are given in Code Availability. A generative AI assistant (Anthropic Claude, via Claude Code) was used during this work for software implementation, data-pipeline construction, statistical tooling, and drafting and editing of manuscript text. This is not AI-assisted copy editing, so it is declared rather than exempt. The methodological decisions the dataset rests on — the pre-registration of the quality rules, the choice to freeze and checksum the splits before any model was run, the leave-city-out protocol, the treatment of informative missingness, and every retraction recorded in `data/DECISIONS.md` — were made by the authors, who verified every reported figure against the regenerated result tables and take full responsibility for the content. See *Use of Generative AI* for the full declaration.

Data Records

The benchmark is deposited as a single versioned archive, **`eco-pulse-ca` v1.1.0**. Every file below is plain text (JSON or CSV). `splits.sha256` carries the digest of `splits.json`, which the test suite verifies on every run; the three split sidecars are projections the builder writes from the same payload and are not separately digested, so `splits.json` is the file to cite.

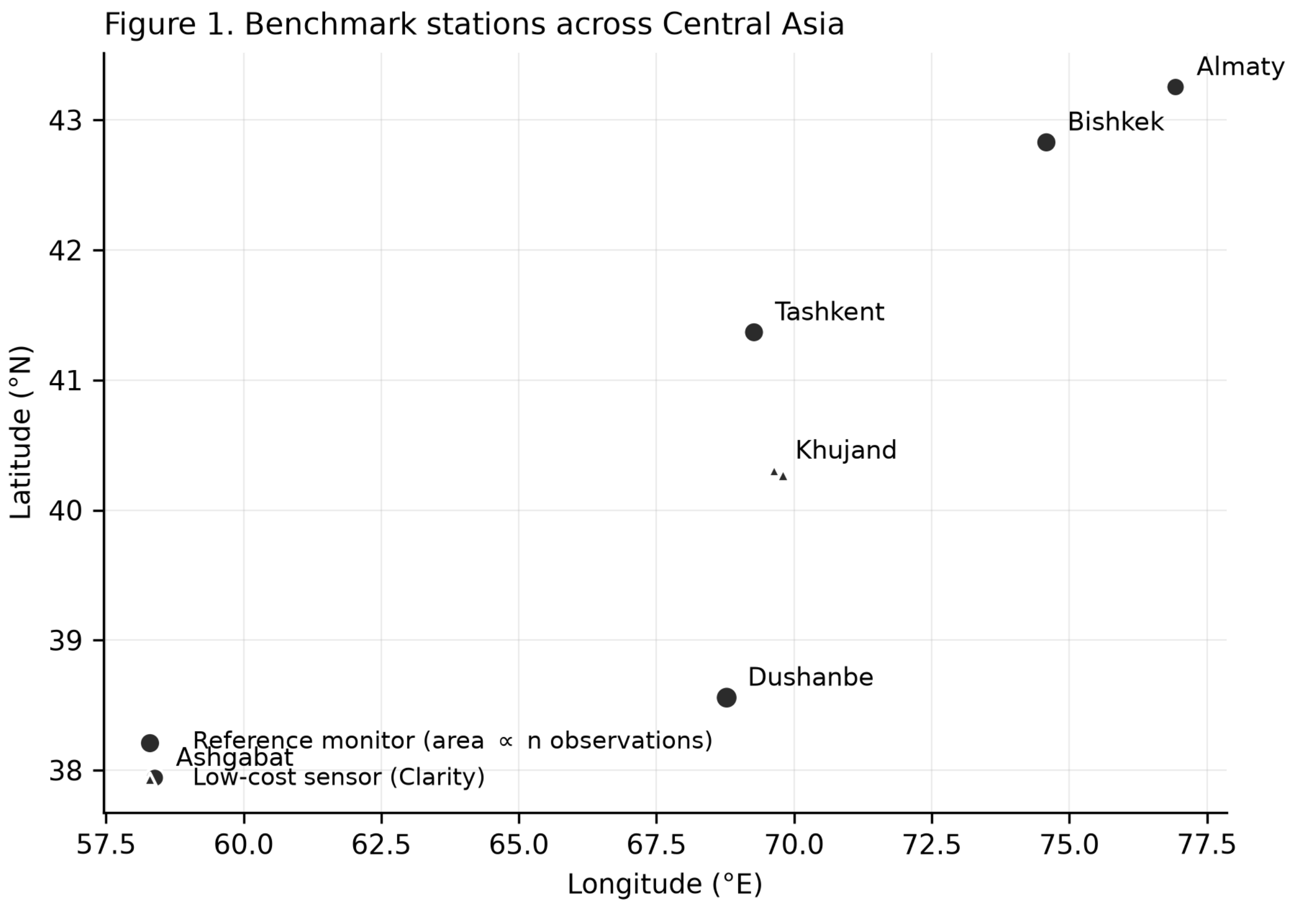

Figure 1. The 7 benchmark instruments across 6 Central Asian cities. Marker style distinguishes the 5 US diplomatic-post reference monitors from the 2 Clarity low-cost sensors at Khujand. Leave-city-out withholds every instrument in one city at a time, so the spacing between cities — not between instruments — sets the extrapolation distance each fold demands.

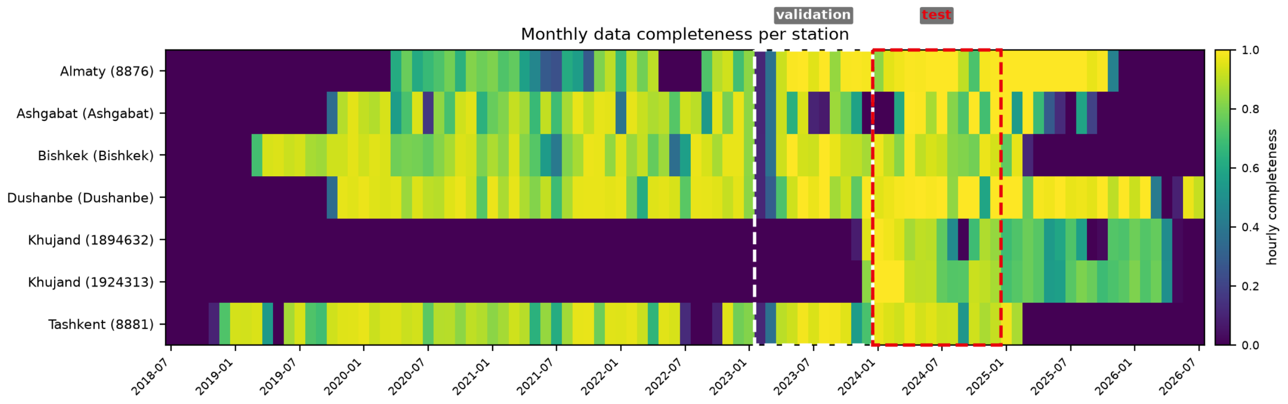

Figure 2. Monthly hourly-completeness for each of the 7 benchmark stations, from first to last observation. Dashed boxes mark the validation (2023-01-11 to 2023-12-21) and test (2024-01-01 to 2024-12-31) blocks. Two features of the record are visible directly: both Khujand sensors begin only in late 2023, after the training block closes, which is what makes Khujand a zero-label fold; and the record ends unevenly, with 2 of 7 stations (8881, Bishkek) stopping at the StateAir closure on 2025-03-04 while the remainder continue past it through a longer-lived feed. All of that lies beyond the test block and none of it is used.

The frozen benchmark definition

These five files *are* the benchmark. A method is evaluated on this benchmark by consuming them; nothing else in the archive is required.

File	Format	Size	Contents
`splits.json`	JSON	4,994 B	The complete benchmark definition: version, station table, temporal blocks, leave-city-out folds, leave-station-out folds, and the protocol configuration. Self-contained.
`temporal_blocks.json`	JSON	725 B	Train, purge, validation, purge, test and reserved blocks with UTC bounds.
`leave_city_out.json`	JSON	1,771 B	The 6 spatial folds, each naming held-out city, held-out stations and training stations.
`leave_station_out.json`	JSON	493 B	The 2 within-city folds, plus the cities named ineligible and why.
`splits.sha256`	text	333 B	SHA-256 of each file above, with the freeze timestamp.

`splits.json` — record structure.

`stations` (7 records). One per instrument: `station_id`, `city`, `latitude`, `longitude`, `n_observations`. Station identifiers are OpenAQ `location_id` values as strings, except where two co-published feeds were merged into one physical instrument, in which case the identifier is the city name (see Methods, and `panel_sources.parquet` for per-hour provenance of the merge).

`temporal_blocks` (6 records). `name`, `start`, `end`, in UTC. The two `purge_*` blocks are excluded from training and evaluation by construction and exist to break autocorrelation across the boundary.

leave_city_out (6 records). fold, held_out_city, held_out_stations, train_stations, n_train_cities.

leave_station_out (2 records plus an ineligible_cities list). fold, city, held_out_station, train_stations.

config. max_lag_hours = 168, max_horizon_hours = 72, purge_hours = 240, test_year = 2024, seeds = 0, 1, 2, 3, 4, and purge_rule, which states the arithmetic relation the purge must satisfy so that a reader can verify it rather than trust it.

Reference results

Provided so that a new method can be placed on the same axes without re-running the reference implementation. All are CSV with a header row.

File	Rows	Contents
t3_01_task_f_baselines_hourly.csv	60	Task F baseline ladder, hourly, by station and horizon.
t3_02_task_n_baselines_hourly.csv	28	Task N baseline ladder, hourly, by station, with exceedance metrics.
t3_06_task_n_baselines_daily.csv	49	**Task N baseline ladder at daily resolution**, scored on the same evaluation rows as the learned models. This is the table to compare a daily model against; the hourly tables are not comparable to it.
t4_01_cams_baseline_variants.csv	21	CAMS raw, locally debiased and pooled-debiased, per station.
t5_01_loco_untuned.csv	108	Untuned gradient boosting, leave-city-out, per fold and seed.
t5_02_loco_tuned.csv	123	Tuned gradient boosting, leave-city-out, per fold and seed, with selected hyperparameters.
t6_01_predictions_task_n.csv	2,214	**Row-level predictions** on the test block: station_id, date, observed pm25, ensemble lgbm, per-seed lgbm_seed0–lgbm_seed4, pooled (debiased CAMS), fold.
t6_02_dm_lgbm_vs_cams.csv	7	Diebold–Mariano per fold and pooled.
t6_06_significance.csv	7	Primary and sensitivity inference with unit of analysis, statistic, *p* and confidence interval.
t6_07_per_fold_holm.csv	6	Per-fold *p*-values with Holm step-down correction.
t7_06_leave_khujand_out.csv	2	**Leave-Khujand-out sensitivity.** The primary inference recomputed over the five reference-grade cities, with each arm's city count, row count, RMSEs, mean loss differential, both *p*-values, the attainable permutation floor and a rule-assigned verdict.

t6_01_predictions_task_n.csv is the most reusable record: it permits any alternative loss, significance procedure or aggregation to be applied to the reference implementation without retraining it, and its per-seed columns make the ensemble decomposable.

What is not redistributed, and why

The derived ground-truth panel is not included. It is built from the OpenAQ archive, whose per-feed licence terms are heterogeneous: six of the ten source feeds carry an explicit licence permitting redistribution and four carry no licence record at all. Depositing the merged panel under a single licence would assert a uniform permission the evidence does not support for every feed. The full matrix, retrieved 2026-08-14, is in `data/MANIFEST.md` and summarised in Data Availability. `data/MANIFEST.md` records the full provenance of every source, and two documented commands rebuild the panel from the archive with an API key.

This is a real limitation of the deposit and is stated rather than worked around: the split definitions and reference results are fully self-contained and verifiable offline, but *regenerating* the reference results from raw observations requires that acquisition step.

Technical Validation

Validation is reported in three layers: the quality control applied to the observations, the integrity of the split definitions, and the behaviour of a reference implementation evaluated under the protocol. The third layer is included because a benchmark whose reference implementation has never been run is not validated — but its results are presented as evidence about the *task*, not as a claim of method performance.

Observation quality control

Seven rules (Q1–Q7) were declared in `data/DECISIONS.md` **before the data were inspected**, each recording its effect on *n* and the direction of bias if the rule is wrong. Two produced findings that changed the benchmark. **One further rule, Q5c, was not pre-registered**: it was added during validation, after the pre-registered rules had passed a station pair that inspection showed to be a single instrument. That sequence is described below rather than presented as foresight.

Duplicate identity — and the case a distance rule cannot see. The US embassy monitors are published twice, by StateAir and by AirNow, under separate identifiers. Where the two records agree on position (Bishkek, 57 m; Ashgabat, 40 m) a 150 m co-location rule catches them. Where they disagree it does not. Dushanbe's two records sit **6.06 km apart** and were initially treated as distinct sites — the manuscript even cited that separation as evidence they were distinct.

They are not. Over the 33,462 hours in which both report, **94.0% of readings are bit-identical**, and of the remainder 99.9% match exactly at a five-hour offset — Dushanbe is UTC+5, so those are the same measurements timestamped in local time rather than UTC. Taken together **99.99% of overlapping hours are the same reading**. The comparable Khujand pair, 14.4 km apart, is bit-identical on 0.3% of hours.

A value-identity rule (Q5c) was therefore added: flag any station pair whose overlapping observations are bit-identical on more than half of samples, regardless of separation. Two independent instruments do not agree to floating point. The threshold sits in the middle of a measured 36× gap between coincidence (2.6% for unrelated station pairs, an artefact of hourly values being reported as rounded integers) and duplication.

The Dushanbe records are merged under the same precedence-and-gap-fill rule already applied to Bishkek and Ashgabat — never averaging, because averaging two copies of one measurement reduces no noise and fabricates a third value where they differ. The benchmark holds **7 instruments across 6 cities**, and the version was raised to 1.1.0 to mark that the splits changed.

Timezone correctness. Metadata offsets are not trusted; each station's diurnal composite is cross-correlated against a regional reference. An initial implementation rejected both Khujand sensors for an apparent 12-hour shift, which investigation showed to be an artefact: the regional reference self-correlates at r = +0.71 under a 12-hour rotation, because Central Asian urban $PM_{2.5}$ is bimodal with peaks roughly half a day apart. The check now reports lag identifiability and flags rather than rejects when the hypothesis cannot be distinguished.

A limitation of this check must be stated. It compares stations against city peers, and only Khujand holds more than one instrument. **A constant, lifelong offset at a single-instrument city is undetectable by any check in this suite.**

Instrument grade

5 of 7 instruments are US diplomatic-post monitors published by AirNow or StateAir (`is_monitor = true` in the OpenAQ census). These are BAM/FEM-class beta-attenuation instruments operated under the programme's documented QA regime, and they are the region's only consistent multi-country reference. **The 2 Khujand instruments are Clarity Node-S low-cost optical sensors** (`is_monitor = false`), which use light scattering rather than gravimetric-equivalent measurement and carry the humidity- and composition-dependent uncertainty documented for that class (Zheng et al., 2018). This matters more than a metadata note, because Khujand is the benchmark's zero-label fold: the city contributes no training row anywhere in the record, making it the harshest spatial test the benchmark contains *and* the one city whose labels carry low-cost measurement uncertainty (Zheng et al., 2018).

Both Khujand instruments also satisfy the pre-registered 2-year span rule only by counting observations after the benchmark record ends; inside the window their spans are 1.09 y and 1.07 y. Admitting them was a coverage decision, and it is recorded as one.

Split integrity

- **Immutability.** `splits.sha256` is compared against a fresh build on every test run. The test fails for the authors exactly as it fails for anyone else; changing a split requires raising the version, recording the reason, and regenerating every published number.
- **Purge sufficiency.** `purge_hours` = 240 = `max_lag_hours` (168) + `max_horizon_hours` (72), verified arithmetically at both boundaries so that no training row's feature window reaches into the evaluation block.
- **Fold disjointness.** Automated assertions verify that no leave-city-out fold trains on any station in its held-out city, that no leave-station-out fold trains on its held-out station, that every evaluated row falls inside the test block, and that each fold evaluates only its own city.
- **Tuning isolation.** Hyperparameters are selected on the validation block; a test parses the tuning function and fails if it references the test block at all.
- **Baseline isolation.** The CAMS bias correction is fitted on the training block only, and the pooled variant excludes the held-out city from its own correction — otherwise the reference every model is measured against would itself contain test information.

Reference implementation behaviour

Scored at a single temporal resolution on the frozen test block, over 5 seeds:

Model (leave-city-out, daily)	RMSE μg/m³
nearest_monitor	33.50
training_pool_mean	32.75
train_global_mean	32.70
train_global_median	30.99
ordinary_kriging	29.75
idw_k5_p2	29.44
LightGBM, retrospective (log target)	**28.01**

The tuned reference model leads every admissible baseline, and explains little within-city variation. It scores 28.01 ± 0.35 µg/m^3 against 29.44 µg/m^3 for the strongest legal rung (`idw_k5_p2`), ahead of all 6 of them **on the fold mean**. That ordering is not robust: the paired difference over 6 cities is not statistically separable (p = 0.586), per-fold differences span -10.73 to +5.10 µg/m^3, and removing one city reverses the lead in one of 6 subsets. The margin is 4.1× the seed standard deviation, so it is fold heterogeneity rather than run-to-run noise (`t7_05_ranking_robustness.csv`). A constant equal to the held-out city's own test-block mean scores 28.12 µg/m^3, but that predictor uses test labels and is **not legal** under leave-city-out; it is reported as a diagnostic floor only. Mean per-fold R^2 is -0.04, with a spread of -0.55 to 0.52 and 3 of 6 folds negative. Pooled against the global mean, R^2 = 0.13. The model captures some between-city variation and within-city skill that does not generalise across cities: per-fold R^2 is positive in 3 of 6 cities and negative in the rest. A reader given only the pooled figure would substantially overestimate what it does.

Baselines are scored at the same daily resolution on the same evaluation rows; an hourly RMSE is structurally larger than a daily one and the two are never compared.

Where the data are hard

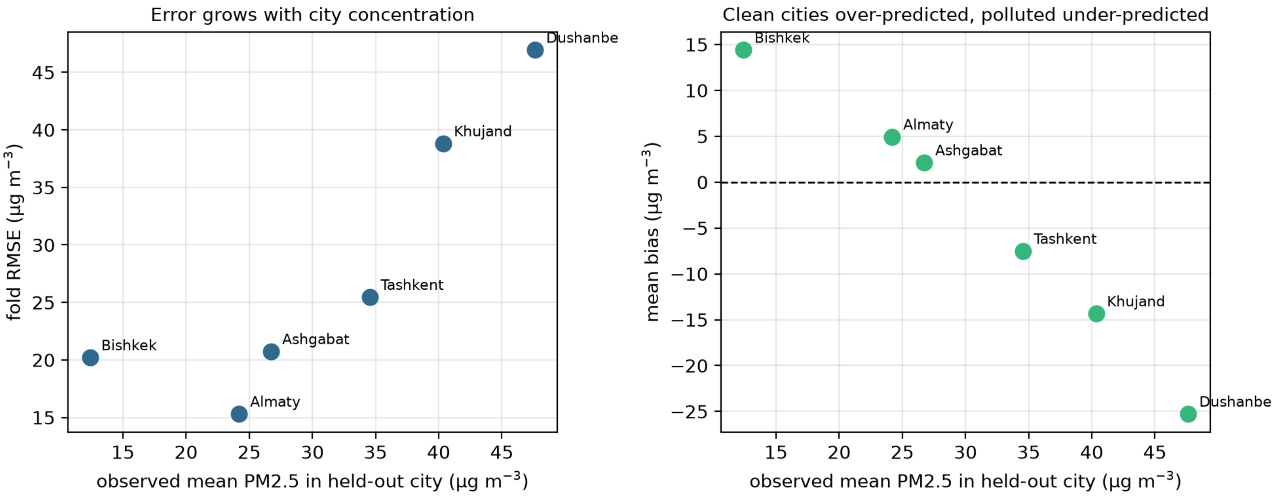

Figure 3. Leave-city-out error against the held-out city's mean PM2.5. Left: fold RMSE rises with city concentration (Spearman rho = 0.94). Right: mean bias falls monotonically from 14.4 µg/m^3 in Bishkek, the cleanest city, to -25.3 µg/m^3 in Dushanbe, the most polluted (rho = -1.00).

The two panels are not equally strong evidence, and the difference matters for reuse. RMSE scales with the variability of whatever is being predicted, so the left panel is partly a scale effect: dividing each fold's RMSE by that city's own observed standard deviation leaves no monotone relation with concentration at all (rho = -0.03). The bias panel is monotone across every fold without exception (rho = -1.00), and the reason is visible in the predictions themselves rather than in the coefficient. The model's predicted city means span only 6.77 µg/m^3, from 22.37 to 29.14, while the observed city means span 35.23 µg/m^3, from 12.42 to 47.64. A predictor that flat is biased upward in clean cities and downward in polluted ones by arithmetic, and a perfect rank correlation over six folds is what that arithmetic produces. The finding is therefore the flatness: transferred to a city it has never seen, the model returns something close to a regional level and does not move it with the city. The same pattern holds within the concentration range: bias is 10.1 µg/m^3 on days below the WHO 24-hour guideline and -90.4 µg/m^3 above six times it, where RMSE reaches 100.9 µg/m^3 on the 6.6% of rows in that band. Winter (DJF) RMSE is 51.0 µg/m^3 against 16.0 µg/m^3 in summer.

This is a property of the region and the network rather than of one model, and it is the clearest thing the dataset shows: **transferring a PM2.5 model to an unmonitored Central Asian city fails first on the city's overall level, not on its day-to-day pattern.** Per-fold, per-band and per-season figures are in `t7_01`–`t7_03`.

Statistical validation

The comparison against bias-corrected CAMS is reported under a primary analysis declared on scientific grounds, with sensitivity analyses, rather than by selecting a favourable test.

The estimand is the reduction in squared error **at a city with no local training labels**, so the unit of generalisation is the city. Aggregating to one value per city gives 6 observations.

The quantity tested is the loss differential Δ, the model's squared error minus debiased CAMS's, in (μg/m³)². Negative values favour the model.

	Test	Unit	*n*	Δ (95% CI)	*p*
Primary	paired *t* on city means	city	6	**-96.2 (-237.0, +44.5)**	**0.1392**
Primary	exact sign-flip permutation	city	6	**-96.2**	**0.1250**
Sensitivity	station-day, independence assumed	station-day	2214	-101.7 (-133.3, -70.2)	2.6e-10
Sensitivity	station-day, Newey–West HAC (lag 60 d)	station-day	2214	-101.7 (-171.7, -31.7)	0.0044
Sensitivity	cluster bootstrap over cities	city	6	-96.2 (-196.5, -2.9)	0.0428

The interval is the more informative half of the primary row. At (-237.0, +44.5) (μg/m³)² it spans zero and is wide enough to contain both a substantial improvement over CAMS and a moderate degradation. The study does not establish that the model is better, and it equally does not establish that it is not: with 6 cities the design cannot separate those cases. Reporting only p = 0.1392 would invite the reading that no effect exists, which the interval does not support either.

The loss differential is serially correlated within station (first-order autocorrelation 0.25), so the independence-assuming figure is reported only for comparison with earlier drafts and is not a valid inference. With 6 clusters, cluster-robust asymptotics are also unreliable — the variance estimator is materially downward-biased below roughly 30–50 clusters (Cameron and Miller, 2015) — which is why an exact permutation test is reported alongside. Its smallest attainable two-sided p-value is 0.03125, a floor imposed by having only 6 cities, stated so that it is not mistaken for evidence.

Per-fold Diebold–Mariano tests are additionally Holm-corrected (Holm, 1979) for 6 comparisons, the family being the per-fold comparisons of one model pair; **3 of 6 survive at α = 0.05.**

The evidence does not support a claim that the reference model outperforms bias-corrected CAMS under the unit of generalisation this benchmark is built around. That is reported as the result.

Evaluation protocol, and a disclosure

The test block was scored under **two** frozen configurations. Both selections were made on the validation block; neither used test performance as a criterion. The history is given in full so a reader can judge it rather than take it on trust.

Freeze 1 — target transform. Daily PM2.5 in this record has skew 2.79 and excess kurtosis 13.5, so squared error on the raw scale is dominated by a few extreme days. Four formulations (raw, log1p, and residual-against-interpolator variants of each) were compared across three model families on validation only. `log1p` won for every family. Frozen, then scored once on test: fold-mean RMSE fell from 30.24 to 28.05 μg/m³.

Freeze 2 — feature exclusion. A validation ablation indicated that satellite retrieval-count features harmed leave-city-out generalisation, which is mechanistically plausible: retrieval success depends on local surface brightness, snow cover and solar geometry, all city-specific. Frozen, then scored once on test: **the validation gain of 1.75 μg/m³ did not replicate, delivering 0.045 μg/m³.**

That non-replication is reported rather than removed. Reverting the configuration after seeing its test result would have made the test set a selection criterion, which is the practice the frozen protocol exists to prevent. The exclusion therefore stands, and the finding — that validation-based feature selection over six cities does not reliably transfer — is itself information for anyone reusing these splits.

No city, station, date or evaluation period was excluded at any point on the basis of its effect on a score. `scripts/experiment_model_search.py` and `scripts/experiment_ablation_ensemble.py` never read the test block, which is enforced by `tests/test_frozen_configuration.py`.

Reproducibility

A single command regenerates every reported number from the frozen splits. Two consecutive runs reproduce all 30 result tables **byte-identically**, verified by SHA-256. The manuscript is rendered from templates whose numeric fields are substituted from a machine-extracted mapping, so no reported figure is typed by hand, and a test re-extracts from the CSVs and fails if any figure has drifted.

The reference implementation is deterministic: the per-seed RMSEs in `t5_02` and the per-seed predictions in `t6_01` agree to floating point across all 6×5 fold-seed pairs, and the ensemble satisfies the convexity bound relating it to its members in every fold.

Usage Notes

How to score a method on this benchmark

1. Read `splits.json`. It is self-contained: station table, temporal blocks, and both fold definitions.
2. Fit only on the rows a fold permits. For leave-city-out, no observation from the held-out city may enter training — including indirectly, through a neighbour feature or a bias correction estimated on that city.
3. Predict on the test block (2024-01-01 to 2024-12-31) and score against the daily target: local-calendar daily mean PM2.5, requiring at least 18 hours of observations.
4. Report **per city, not pooled**, with dispersion across at least the 0, 1, 2, 3, 4 seeds.
5. Compare against `t3_06_task_n_baselines_daily.csv`, which is scored on the same rows at the same resolution.

Six rules the benchmark asks of every submission

These are the reporting conditions under which numbers from different methods remain comparable. They are applied to the reference implementation in this paper as well.

1. **State the task.** Task N (nowcasting at unmonitored sites) and Task F (forecasting at monitored stations) are different problems with different admissible features. Never pool them, and never quote a Task F number as a general accuracy.
2. **State the tier.** `static_only`, `deployable` and `retrospective` differ in whether a predictor could exist at prediction time. A retrospective number is not a deployment claim.
3. **Report per city.** A pooled figure over 6 cities of unequal size hides which cities fail.
4. **State the truncation lag** used in any Diebold–Mariano test, and report a sensitivity sweep. A verdict that flips with the lag choice is not a verdict.
5. **Report seed dispersion.** At least mean and standard deviation over the declared seeds.
6. **State the unit of analysis** for any significance claim, and justify it. Station-days are not independent observations.

Pitfalls this benchmark is designed to expose

Temporal resolution is not a free choice. RMSE computed on hourly observations is structurally larger than RMSE on daily means of the same data, because averaging removes within-day variance. An hourly baseline placed beside a daily model produces an apparent margin that is arithmetic, not skill. This paper made that error and corrected it; the daily ladder exists so that the comparison is unambiguous.

Two aggregations of R^2 answer different questions. Mean per-fold R^2, computed against each city's own mean, measures within-city day-to-day skill. Pooled R^2 against the global mean largely measures whether cities differ from each other, which is far easier. Report which one you mean. On this benchmark the reference model scores -0.04 on the first and 0.13 on the second.

Exceedance F1 has a high floor. 4 of the 6 cities clear the WHO 24-hour guideline on most days, from 88% of test days at Dushanbe down to 24% at Bishkek, so a classifier that always predicts "exceeds" scores F1 = 0.741 at a base rate of 61.8%. Peirce skill score is reported alongside because it is zero for that classifier by construction.

A single station can move the conclusions. Merging one duplicated instrument changed every fold's RMSE and reversed the ordering of feature-attribution families. Attribution rankings on a 7-instrument benchmark should be treated as unstable, and this paper's own earlier claim about them is retracted in Technical Validation.

One city carries 26.7% of the pooled rows. Khujand is the only two-station city, so any row-level statistic here is weighted toward it, and it is also the only city with low-cost labels. `t7_06_leave_khujand_out.csv` recomputes the primary inference without it: the verdict is unchanged (ROBUST), with paired *t* $p = 0.1392$ over all 6 cities against 0.2165 over the five reference-grade ones. Report per city, and if you must pool, run the same exclusion.

Known limitations

- **Six cities.** The spatial sample is small, and it bounds every inference: with 6 clusters, no significance procedure can attain a two-sided p below 0.03125 without distributional assumptions the data do not support.
- **Two of 7 instruments are low-cost**, and they constitute the zero-label Khujand fold. Results for that fold are not comparable in kind to the other five.
- **Single-instrument cities cannot be timing-audited.** Only Khujand holds two genuinely distinct instruments, so a constant lifelong offset at Almaty, Tashkent, Bishkek, Ashgabat or Dushanbe would be invisible to every check in the suite. Dushanbe belongs on that list only after the merge recorded in `DECISIONS.md` D-012, which established that its two records were one instrument published twice.
- **The record ends before the source does.** Every StateAir feed stops on 2025-03-04, when that publication channel closed, and the **evaluated** record ends with the 2024 test block. Observations after it exist in the panel but are reserved and unused. The monitors themselves did not all stop: as of 2026-08-14 the same US diplomatic-post instruments are still republished through AirNow at Ashgabat (to 2025-09-24), Almaty (to 2025-11-14) and Dushanbe (still reporting), while Bishkek, Tashkent and Astana ceased on 2025-03-04. No result here speaks to current conditions, but the record *can* be extended forward for the three cities whose AirNow feed continued — doing so would change the benchmark and requires a version bump, not a silent refresh.
- **Kazakhstan contributes one city.** Astana failed the completeness rule at 42.8% against a required 60%.
- **Task F is single-horizon.** The learned Task F model predicts next-day daily means and does not produce the 48 h or 72 h horizons the hourly Task F ladder in `t3_01` covers; the two are not comparable and are not tabled together.
- **The ground-truth panel is not redistributed.** Split definitions and reference results are self-contained and verifiable offline. Regenerating results from raw observations requires rebuilding the panel from the OpenAQ archive, whose per-location records are downloadable anonymously from its open-data bucket on Amazon S3; the v3 API route additionally needs a free key.

Intended reuse

The benchmark is designed for methods comparison, not for producing exposure estimates. Any method that yields a PM2.5 estimate — statistical, physical or learned — can be scored on these splits and placed against the baseline ladder without re-deriving a protocol. Because `t6_01_predictions_task_n.csv` banks row-level and per-seed predictions, an alternative loss function, aggregation or significance procedure can be applied to the reference implementation without retraining it.

Data Availability

The frozen benchmark — split definitions, checksums and reference result tables — is deposited in Zenodo at **https://doi.org/10.5281/zenodo.21930669** under a CC BY 4.0 licence, as version 1.1.0 of `eco-pulse-ca`. The archive contents and record structure are described in Data Records.

The identifier above is the **version** DOI, which resolves to this release specifically rather than to the latest version. Results reported against this benchmark should cite it, together with the `splits.sha256` checksum, so that a score is attributable to one frozen split definition.

What the deposit does and does not contain. The archive holds the benchmark's *derived* artefacts: split definitions, checksums, reference result tables, row-level predictions and the full pipeline code. **It does not contain the underlying hourly PM2.5 observations**, and the split definitions are not a substitute for them — they are station identifiers, fold membership and time bounds, not measurements.

Ground PM2.5 observations are accessed through the OpenAQ archive (openaq.org) and originate with the US Department of State AirNow and StateAir programmes and with Clarity.

Licence status, verified 2026-08-14. Per-location licence records were retrieved from OpenAQ's `/v3/locations` endpoint and are tabulated in full in `data/MANIFEST.md`. Six of the ten source feeds carry an explicit licence permitting redistribution: the four AirNow feeds are recorded as *US Public Domain* and the two Clarity feeds as *CC0 1.0*, both with redistribution and modification allowed and attribution not required. These six account for 63.6% of contributing observation-hours. **The four StateAir feeds (Ashgabat, Bishkek, Dushanbe, Tashkent) carry no licence record at all** — the field is null, and the absence is systematic: across all 33 StateAir providers, none of their 36 locations carries one. The absence tracks the provider *label* rather than the data. OpenAQ's issue tracker documents a single location ingested under both the AirNow and StateAir labels, alternating between them, and that location today carries the AirNow label and a US Public Domain licence; the licence follows the label. We therefore treat the null as unassigned metadata rather than a withheld permission, while recording that no licence record has been issued for these four feeds.

The Department of State's archived *Data Use Statement* for its air-quality programme requires that products relying on the data give attribution to the Department and indicate that the data "are not fully verified or validated", and asks that the data "not be altered in any way and be disseminated as received". Its stated scope is the Mission China programme and the `stateair.net` portal, neither of which served these four Central Asian feeds, and the portal that did serve them published no equivalent statement. We nonetheless attribute the observations to the U.S. Department of State and carry the verification caveat, and we treat the final clause as a further reason not to redistribute a merged, quality-controlled, daily-aggregated panel. `data/MANIFEST.md` sets out the full evidence and its limits.

Predictor sources. Satellite retrievals are Sentinel-5P (CO, NO_2, SO_2, absorbing aerosol index) and MODIS MAIAC AOD; the chemistry-transport forecast is Copernicus CAMS. ERA5 reanalysis was retrieved and latency-tested but enters no feature tier (Methods), so it is a source the pipeline touched rather than a predictor the results use. Each source, its access route, its licence status and its measured acquisition latency are recorded in `data/MANIFEST.md`.

Why the panel is not deposited here. Licence terms are heterogeneous across the ten source feeds, and for the four StateAir feeds no licence has been recorded by the platform that serves them. Depositing the merged panel under a single licence would assert uniform permission that the evidence does not support for every feed, so the observations are left at source. This is a statement about provenance, not about availability.

Every observation used here is publicly retrievable, without credentials. All ten source feeds — the four StateAir feeds included — are published in OpenAQ's open-data archive on Amazon S3 (`s3://openaq-data-archive/records/csv.gz/locationid={id}/`), registered with the AWS Registry of Open Data and downloadable anonymously; retrieval of each benchmark feed was verified there on 2026-08-14. The same records are available through the OpenAQ v3 API with a free key. `data/MANIFEST.md` lists every source location identifier, and two documented commands rebuild the panel. The split definitions and reference results are fully self-contained and verifiable without any of this.

Code Availability

All code used to build the benchmark, run the reference implementation, produce every table and figure, and render this manuscript is openly available at `https://github.com/jalencik/eco-pulse-benchmark`, archived at the deposit above, under an MIT licence.

Environment. Python 3.12 (`>=3.12,<3.13`), with numpy, pandas, scipy, scikit-learn, LightGBM, pyarrow and matplotlib. `pyproject.toml` declares compatible ranges; `requirements-lock.txt` records the exact versions the deposited tables were produced with, and is the file to install from when byte-identical regeneration is the goal. `python tasks.py setup` creates the environment.

Reproduction. A single command runs the whole chain — lint, type check, the full test suite, checksum verification, split regeneration, baseline ladder, model layer, table regeneration and manuscript rendering:

```
make reproduce        # or, where make is unavailable:  python tasks.py reproduce
```

The command is deterministic: two consecutive runs reproduce all 30 result tables byte-identically under SHA-256. Verifying the frozen splits requires neither credentials nor a rebuild (`sha256sum -c splits.sha256`), and the test suite runs offline against committed fixtures. Regenerating the reference results additionally requires the ground-truth panel described above.

Custom code of note. The quality-control rules (`src/ecopulse_ca/qc/`), split builder (`src/ecopulse_ca/splits/`), Diebold–Mariano implementation with Harvey–Leybourne–Newbold correction (`src/ecopulse_ca/eval/`), and the primary/sensitivity inference (`scripts/build_significance.py`) are original to this work. 592 automated tests enforce split immutability, absence of leakage, table provenance and manuscript-number consistency.

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Author Contributions

Jaloliddin Musayev: Conceptualisation; Methodology; Software; Validation; Formal analysis; Investigation; Data curation; Writing — original draft; Writing — review and editing; Visualisation; Project administration.

Asadbek Abdivayitov: Data curation; Investigation.

Ozodbek Yo'ldashev: Supervision; Writing — review and editing.

All authors read and approved the submitted manuscript.

Competing Interests

The authors declare no competing interests.

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Use of Generative AI

During the preparation of this work the corresponding author used Anthropic Claude (Claude Code) to assist with software implementation, data-pipeline construction, statistical tooling, and drafting and editing of the manuscript text. The authors reviewed and edited all output and take full responsibility for the content. Generative AI is not listed as an author and holds no accountability for this work.

Every reported figure is machine-extracted from the banked result tables and substituted into the manuscript at build time; an automated test re-extracts from the source CSVs and fails if any figure has drifted. Prose-level editing performed with AI assistance is recorded openly in the repository's commit history.

Data and Code Citation

Users of this benchmark should cite the archived version and its checksum, not a version-control branch head, so that a reported score is attributable to a specific frozen split. The benchmark version described here is **1.1.0**.